CHAPTER 11

THREE-DIMENSIONAL GEOMETRY

CHAPTER ESSENCE AT A GLANCE:
Three-dimensional geometry, direction cosines, vector and cartesian equations of lines in space, and shortest distance between skew lines.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Lines in Three Dimensions:** A straight line in three-dimensional space is uniquely determined by a given point and a direction vector, represented equivalently in parametric vector notation or symmetric Cartesian ratios.

• **Shortest Distance Between Skew Lines:** Non-intersecting and non-parallel spatial lines are skew lines; the shortest distance between them is measured along their common perpendicular, evaluated using scalar triple products.

• **Distance Between Parallel Lines:** The shortest separation between two parallel spatial lines is determined by the cross product of their shared direction vector with the displacement vector connecting points on each line.

• **Direction Ratios and Perpendicularity:** Two lines in three dimensions are perpendicular when the sum of products of their corresponding direction ratios equals zero, and parallel when their direction ratios are in constant proportion.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{b} \Leftrightarrow \frac{x - x_1}{a_1} = \frac{y - y_1}{b_1} = \frac{z - z_1}{c_1}$$

Vector and Cartesian symmetric equations of a line in 3D

$$d = \frac{|(\mathbf{b}_1 \times \mathbf{b}_2) \cdot (\mathbf{a}_2 - \mathbf{a}_1)|}{|\mathbf{b}_1 \times \mathbf{b}_2|}$$

Shortest distance formula between two non-parallel skew lines

$$d = \frac{|\mathbf{b} \times (\mathbf{a}_2 - \mathbf{a}_1)|}{|\mathbf{b}|}$$

Shortest distance between two parallel spatial lines

$$\cos\theta = \frac{|a_1 a_2 + b_1 b_2 + c_1 c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Acute angle between two lines with given direction ratios

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** [Trap Alert: Cartesian Standard Form] In Cartesian line equations, the coefficients of x, y, z MUST be +1! If given (1 - x) over 2, rewrite as (x - 1) over (-2) so direction ratio is -2, not 2.

• **Alert:** [Trap Alert: Absolute Value Distance] Distance cannot be negative; always apply outer modulus |...| to the scalar triple product numerator.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Find direction cosines of a line that makes equal angles with coordinate axes. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Equal Angles with Axes]** Let line make angles α, β, γ with x, y, z axes. Given $\alpha = \beta = \gamma$. Direction cosines: l = cos α, m = cos α, n = cos α. [1 Mark]

• Identity: $l^2 + m^2 + n^2 = 1$ implies $\cos^2 \alpha + \cos^2 \alpha + \cos^2 \alpha = 1$ implies $3 \cos^2 \alpha = 1$ implies cos $\alpha = \pm 1$ over sqrt(3). Direction cosines are ± 1 over sqrt(3), ± 1 over sqrt(3), ± 1 over sqrt(3). [1 Mark]

• **[Boxed Result: Direction Cosines: (± 1 over √3, ± 1 over √3, ± 1 over √3)]**

2. If a line has direction ratios -18, 12, -4, find its direction cosines. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Direction Cosines from Ratios]** Direction ratios: a = -18, b = 12, c = -4. $\sqrt{(a^2 + b^2 + c^2)} = \sqrt{((-18)^2 + 12^2 + (-4)^2)} = \sqrt{(324 + 144 + 16)} = \sqrt{484} = 22$. [1 Mark]

• l = a over 22 = - 18 over 22 = - 9 over 11. m = b over 22 = 12 over 22 = 6 over 11. n = c over 22 = - 4 over 22 = - 2 over 11. [1 Mark]

• **[Boxed Result: Direction Cosines: -9 over 11, 6 over 11, -2 over 11]**

3. Find vector and Cartesian equations of the line passing through point (5, 2, -4) and parallel to vector 3i + 2j - 8k. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Vector & Cartesian Equations]** Point A(5, 2, -4) $\Rightarrow$ a = 5 hat(i) + 2 hat(j) - 4 hat(k). Parallel vector b = 3 hat(i) + 2 hat(j) - 8 hat(k). [1 Mark]

• Vector equation: r = a + λ b = (5 unit vector i + 2 unit vector j - 4 unit vector k) + λ (3 unit vector i + 2 unit vector j - 8 unit vector k). [1 Mark]

• Cartesian equation: $(x - x_1)$ over a = $(y - y_1)$ over b = $(z - z_1)$ over c: (x - 5) over 3 = (y - 2) over 2 = (z + 4) over -8. [1 Mark]

• **[Boxed Result: Vector: r = (5i+2j-4k) + λ(3i+2j-8k) | Cartesian: (x-5) over 3 = (y-2) over 2 = (z+4) over (-8)]**

4. Find vector and Cartesian equations of line passing through two points (3, -2, -5) and (3, -2, 6). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Line Through Two Points]** Points A(3, -2, -5) and B(3, -2, 6). Direction ratios: $x_2 - x_1 = 3 - 3 = 0$; $y_2 - y_1 = -2 - (-2) = 0$; $z_2 - z_1 = 6 - (-5) = 11$. [1 Mark]

• Vector equation: r = (3 unit vector i - 2 unit vector j - 5 unit vector k) + λ (0 unit vector i + 0 unit vector j + 11 unit vector k) = (3 unit vector i - 2 unit vector j - 5 unit vector k) + μ unit vector k. [1 Mark]

• Cartesian equation: (x - 3) over 0 = (y + 2) over 0 = (z + 5) over 11 (meaning x = 3, y = -2, z is free). [1 Mark]

• **[Boxed Result: (x - 3) over 0 = (y + 2) over 0 = (z + 5) over 11 (Parallel to z-axis)]**

5. Find angle between lines r = (3i + 2j - 4k) + λ(i + 2j + 2k) and r = (5i - 2j) + μ(3i + 2j + 6k). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Angle Formula]** Direction vectors: b_1 = hat(i) + 2 hat(j) + 2 hat(k) and b_2 = 3 hat(i) + 2 hat(j) + 6 hat(k). cos $\theta = |b_1 \cdot b_2|$ over $(|b_1| |b_2|)$. [1 Mark]

• b1 · b2 = 1(3) + 2(2) + 2(6) = 3 + 4 + 12 = 19. |b1| = sqrt(1 + 4 + 4) = sqrt(9) = 3. |b2| = sqrt(9 + 4 + 36) = sqrt(49) = 7. cos θ = 19 over (3 × 7) = 19 over 21 implies $\theta = \cos^{-1}$(19 over 21). [2 Marks]

• **[Boxed Result: $\theta = \cos^{-1}$(19 over 21)]**

6. Derive expression for shortest distance between two skew lines $\vec{r} = \vec{a_1} + \lambda \vec{b_1}$ and $\vec{r} = \vec{a_2} + \mu \vec{b_2}$. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Derivation of Shortest Distance between Skew Lines]** Lines L_1: $\vec{r} = \vec{a_1} + \lambda \vec{b_1}$ and L_2: $\vec{r} = \vec{a_2} + \mu \vec{b_2}$. Let line of shortest distance be PQ perpendicular to both lines. PQ is parallel to $\vec{b_1} \times \vec{b_2}$. Unit vector along PQ is $\hat{n} = (\vec{b_1} \times \vec{b_2})$ over $|\vec{b_1} \times \vec{b_2}|$. [2 Marks]

• Shortest distance d is the projection of vector (a2 - a1) along unit vector n: d = |(a2 - a1) · unit vector n| = | [(b1 × b2) · (a2 - a1)] over |b1 × b2| |. [2 Marks]

• **[Boxed Result: $d = | [(\vec{b_1} \times \vec{b_2}) \cdot (\vec{a_2} - \vec{a_1})]$ over $|\vec{b_1} \times \vec{b_2}| |$]**

7. Find shortest distance between lines $\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$. [Score: 5 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Identify Vectors]** $\vec{a_1} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b_1} = \hat{i} - \hat{j} + \hat{k}$. $\vec{a_2} = 2\hat{i} - \hat{j} - \hat{k}$, $\vec{b_2} = 2\hat{i} + \hat{j} + 2\hat{k}$. $\vec{a_2} - \vec{a_1} = \hat{i} - 3\hat{j} - 2\hat{k}$. [1.5 Marks]

• **[Step 2: Cross Product & Dot Product]** $\vec{b_1} \times \vec{b_2} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 2 & 1 & 2 \end{vmatrix} = \hat{i}(-2 - 1) - \hat{j}(2 - 2) + \hat{k}(1 - (-2)) = -3\hat{i} + 0\hat{j} + 3\hat{k}$. $|\vec{b_1} \times \vec{b_2}| = \sqrt{((-3)^2 + 0^2 + 3^2)} = \sqrt{(9 + 9)} = \sqrt{18} = 3\sqrt{2}$. $(\vec{b_1} \times \vec{b_2}) \cdot (\vec{a_2} - \vec{a_1}) = (-3)(1) + 0(-3) + 3(-2) = -3 + 0 - 6 = -9$. [2 Marks]

• **[Step 3: Shortest Distance]** $d = |-9|$ over $(3\sqrt{2}) = 9$ over $(3\sqrt{2}) = 3$ over $\sqrt{2} = 3\sqrt{2}$ over 2 units. [1.5 Marks]

• **[Boxed Result: Shortest distance $d = 3$ over $\sqrt{2}$ units]**